

FYNZA

Advanced Steganography & Steganalysis Suite

Military-Grade Encryption | AI-Powered Detection | Forensic Analytics

Prepared by the FYNZA Development Team

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1. Executive Summary

FYNZA is a production-grade cybersecurity application designed for secure data transmission and digital forensics. The platform provides a dual-layer security model: encryption and steganography. By hiding encrypted data within multimedia files (Images, Audio, Generic Files), FYNZA ensures that the very existence of the communication is concealed from unauthorized observers.

2. System Architecture

- Backend: Python / Flask API for high-performance processing.
- Frontend: Vanilla HTML5/CSS3/JS with Glassmorphism UI design.
- Plugins: Modular steganography system supporting LSB, DCT, DWT, and more.
- AI Engine: ResNet-18 Deep Learning model for suspicious pattern detection.
- Forensics: Automated PDF reporting and multi-layer heatmap visualization.

3. Steganography Algorithms

The system employs multiple advanced algorithms to hide data:

1. LSB (Least Significant Bit): The foundational method, optimized with 'Camouflage' noise injection.
2. DCT (Discrete Cosine Transform): Hides data in frequency coefficients, making it robust against compression.
3. DWT (Discrete Wavelet Transform): Uses wavelet sub-bands for high-security embedding.
4. SVD (Singular Value Decomposition): Modifies singular values in the image matrix for maximum stability.
5. Adaptive Edge: A scatter-based method that hides data in high-variance regions (noise-like areas).
6. LBB (Local Binary Block): A block-based approach that compares neighbor pixels for embedding.
7. Overlay: Robust watermarking using additive or multiplicative blending formulas.

4. Cryptographic Implementation

Before embedding, every payload is protected by military-grade encryption:

- AES-256-GCM: Provides both confidentiality and authenticity (MAC).
- ChaCha20: A high-speed stream cipher alternative.
- PBKDF2/Scrypt: Used for secure key derivation from user passwords with high iteration counts.
- ZLIB: High-level compression applied before encryption to minimize payload size.

5. AI Steganalysis Engine

FYNZA includes a powerful defense mechanism against steganography. The detection engine uses a ResNet-18 Neural Network trained on millions of original and stego-images. It can identify subtle pixel perturbations and estimate the probability of hidden content, providing a 'Suspicion Score' and identifying potential embedding methods.

6. Forensic Reporting & Analytics

Analysts can use the Forensics Tab to perform deep inspection:

- Change Map: Highlights exactly which pixels were modified.
- Density Map: Visualizes the spatial distribution of the payload.
- Risk Map: Categorizes areas as Safe, Medium, or Risky based on statistical variance.
- PDF Export: Generates a professional legal-ready forensic report including all metrics and heatmaps.

8. The Team

This project was developed by the following contributors:

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